Ritwik Agarwal

CONTACT

919456809610

aritwik93@gmail.com

Portfolio

ritwik-agarwal97

EDUCATION

International Institute of Information Technology (IIIT) Hyderabad

Bachelor of Technology, Building Sciences
2015 - 2019

Master of Science by Research
2024-2025
(Thesis - Indoor Air Quality)

FELLOWSHIPS

- Google Open Source Mentor
- Open Knowledge Foundation Fellow, Frictionless Data
- UCLA Open Scholarly Communications Fellow
- PREreview Champion
- GrowthX

PROFILE SUMMARY

Sustainable Energy / Air Quality Researcher, and Product Manager with 4+ years of experience advancing open science, AI ethics, and tech-for-good initiatives. Current research focuses on Indoor air quality, occupant behavior, and public health.

RESEARCH EXPERIENCE

Research Fellow 2025
Indorama Ventures Center for Clean Energy (IVCEE)

- Prototyped schedules to balance aerosol exposure, integrating occupant behavior and outdoor PM2.5 signals into window-operation logic; building-codes that fuses comfort, air-quality data.
- Leading a field study of Asian bedrooms showing nighttime CO_2 often reach ~1600 ppm with split vs ~1300 ppm with window AC; highlighting ventilation–energy trade-offs in real homes (Accepted to ASHRAE Winter Conference).
- Executed community field deployments: installed T/RH sensors and cool roofs in low-income homes to assess climate-driven urban heat mitigation.

M.S. by Research 2024
Centre for IT in Building Science

- Conducted field household interviews and discrete-choice experiments; found low baseline uptake of purifier plans and modest mean WTP for public clean air programs - linking measured IAQ awareness with barriers to adoption.
- Short-horizon energy forecasting for homes and occupancy detection using transfer learning and neural networks (to couple controls with exposure-aware operation).

RESEARCH & PUBLICATIONS

Energy and Indoor Air Quality Tradeoffs of Personal Air Cleaners and Natural Ventilation in Urban Households

- Published in Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (https://doi.org/10.1145/3736425.3772359)
- Links public willingness to pay with building energy to benchmark IAQ interventions.

Exploring Energy Implications in a Mixed-Mode Building by Linking Air Quality and Occupant Behaviour

- Published in Proceedings of Building Simulation 2025: 19th Conference of IBPSA (https://doi.org/10.26868/25222708.2025.1980)
- Investigates the relationship between occupant behavior, energy consumption, and air quality in sustainable building design.

Behavioral and Socio-Economic Dimensions of Indoor Air Quality: Insights into Public Perceptions and Interventions in India

- Published in Proceedings, Healthy Buildings 2025 (An ISIAQ flagship)
- Explores factors impacting air quality in India, based on household interviews and proposes community-led intervention strategies to improve public health.

A tale of two'opens': intersections between Free and Open Source Software and Open Scholarship

- Link to the manuscript
- Examines the overlapping histories, practices, and principles of FOSS and Open Scholarship

WORK EXPERIENCE

Product Manager 2024
Adri AI | SanFrancisco

- Improved internal processes and outreach processes using in house LLMs which led to a 30% increase in response rates.
- Created engaging videos, product demos and newsletters to demystify complex AI products, helping both technical and non-technical stakeholders understand product value and features.

Product Manager 2023
Kratos Gaming Network

- Directed cross-functional teams across India and Europe to manage partnerships and align game development initiatives, generating 25k on-chain coin transactions monthly and ~100 UAWs daily.
- Designed reward-based systems, including Gacha mechanics, to increase user engagement and promote equitable access to gaming rewards.

Associate Product Manager 2021 - 2023
A23 Games

- Managed notification systems and designed engaging content and graphics, resulting in a 5% increase in community interaction and a .2 rating boost in user satisfaction.
- Launched an ML-based contest recommendation system, improving user retention by 8% and increasing ARPU by 5%.
- Managed third-party collaboration to launch an 8-ball pool game, boosting daily active users (DAU) by 2%.

PROJECTS

Visvayu — Climate Application for Public Health View Project

Collaborated with TU Munich | 2023

- Co-developed an AQI application addressing air quality challenges in Munich, Germany.
- Led community engagement and public outreach for an open-source AQI app, facilitating collaboration between scientists and local residents.
- Organized workshops and feedback sessions to ensure accessible, inclusive design and promote environmental justice.

Blind — Customer Engagement & Retention View Project

- Developed segmentation frameworks to enhance customer engagement and retention on professional networks.
- Redesigned notifications and content strategy, making community interactions more inclusive, relevant, and accessible for diverse professional users.